Hyperspectral Image Classification Based on a Locally Enhanced Transformer Network

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***Abstract*— Recently, transformer-based models have achieved remarkable performance in hyperspectral image (HSI) classification. However, due to the limited training data, existing methods often show limited capability of capturing fine-grained local features. Although attempts have been made to solve this problem, the large amount of parameters imposes the risk of overfitting. In this article, we propose a locally enhanced transformer network for HSI classification with fewer network parameters, which mainly consists of a multibranch spatial–spectral tokenization (MSST) module and a dual-branch transformer encoder (DTE) module. The MSST generates effective spatial–spectral tokens through diverse convolutions with a residual connection. The DTE consists of a global transformer branch and a locally enhanced transformer branch, which are used to capture the global and local spatial dependencies of HSI, respectively. Unlike the conventional self-attention (SA) module used in the global branch, we propose an improved multihead SA (IMSA) module in the local branch by incorporating the local prior information of HSI with graph convolution, to enhance the local information extraction. To fuse the global and local features from the two branches, we introduce an adaptive strategy by using learnable weights for both branches. We devise our MSST and DTE with a shallow architecture, significantly reducing the number of parameters. Experimental results on benchmark datasets demonstrate that the proposed method outperforms the state-of-the-art.**

***Index Terms*— Classification, convolutional neural network (CNN), hyperspectral image (HSI), remote sensing, transformer.**

I. Introduction

HYPERSPECTRAL image (HSI) records the spectral information of land covers in hundreds of bands. The rich spectral information of HSI allows far better recognition of land covers compared with the conventional color or multispectral images [1], [2]. Thus, HSI has a wide range of applications, including mineral detection [3], environmental monitoring [4], and medical diagnosis [5]. In the interpretation of HSI, classification is a fundamental task.

Early HSI classification methods are mostly designed with conventional classifiers, including K nearest neighbors (KNNs) [6], support vector machine (SVM) [7], random forest (RF) [8], and sparse representation classification (SRC) [9], [10]. The performance of these conventional methods highly relies on the hand-crafted features. According to the used features, they can be classified into spectral-based [11], [12] and spatial–spectral-based methods [13], [14], [15]. Spectral-based methods perform classification by using the raw spectrum of HSI or the transformed features of the spectral signature, which only considers the spectral information of HSI. Due to the high dimension of HSI, directly using the raw spectrum often suffers from curse-of-dimensionality. In contrast, the classification methods with transformed features first perform dimensionality reduction by feature extraction or feature selection [16], [17], including principal component analysis (PCA) [18], linear discriminant analysis (LDA) [19], band selection, etc., and then classify HSI with the features of low-dimensionality. The transformed features are often more discriminative, yielding thereby improved performance compared with that using the raw spectrum. However, as only spectral information is exploited, spectral-based methods are often sensitive to outliers or noise, resulting in degraded performance. Thus, spatial-spectral-based methods are proposed by incorporating the spatial and spectral information of HSI. Representative methods include pre-processing based [20] and post-processing-based approaches [21]. The former methods rely on the well-extracted spatial–spectral features by using techniques like Gabor filtering [22], morphological profile [23], etc., before performing classification. While the latter methods typically apply filters to the initially obtained results to improve the spatial homogeneity [24]. These methods yield significant improvement over the spectral-based methods. However, the conventional HSI classification methods typically can only extract shallow features of the data, leading to limited capability in dealing with HSI that often has a highly complex data structure.

In recent years, deep learning methods have made significant breakthroughs in computer vision and high-dimensional data analysis. Compared to traditional feature extraction methods, deep learning-based methods can automatically learn more effective features from data, which facilitates data analysis in various domains [25]. Deep learning-based HSI classification methods have been developed by using neural

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networks, including recurrent neural networks (RNNs) [26], generative adversarial networks (GANs) [27], and graph convolutional networks (GCNs) [28], which yield remarkable performance improvement compared with the traditional shallow classification methods. Among them, the convolutional neural network (CNN) is the most popular network in the field due to its efficiency and powerful capability in extracting local spatial information of HSI. Representative CNN-based classification models include the 1D-CNN [29], 2D-CNN [30], and 3D-CNN [31], which focus on the extraction of spectral feature, spatial feature, and spatial–spectral feature, respectively. The hybrid networks [32] combine 2D-CNN and 3D-CNN and yield improved performance compared with the model using 3D-CNN alone. In order to effectively extract deeper features, skip-connected CNNs are introduced for HSI classification, including the ResNet [33] and DenseNet [34], which reduce the loss of information from the shallow layers. Dong et al. [35] use gray-level co-occurrence matrices to improve the spatial feature extraction. Motivated by the attention mechanism, a number of attention-based CNN neural networks are developed [36], [37], [38]. Zhu et al. [39] combine spectral attention and spatial attention to reweight the features in all channels and spatial locations, allowing the network to adaptively extract more informative features for HSI classification.

Despite their excellent performance, the CNN-based classification models are limited by the restricted receptive field of convolution, especially with a small kernel size, making the long-range information extraction difficult. Recently, vision transformer (ViT) [40] has achieved impressive performance in HSI classification. Benefiting from the self-attention (SA) mechanism [41], the receptive field of ViT is significantly increased compared to CNN, which facilitates effectively capturing the long-range contextual information of data. Recent transformer-based HSI classification methods include [42], [43], [44], [45], [46], [47], which enhance the performance of conventional ViT from the aspects of improving tokenization or attention mechanisms. Hong et al. [48] group adjacent bands to generate input tokens, which are then fed to the encoder. FUST [49] feeds flattened HSI patches into a transformer branch and captures the long-range dependencies of spectral bands. Instead of generating tokens directly from the raw HSI, many studies adopt a hybrid CNN-transformer architecture to better capture the spatial–spectral information of HSI. Sun et al. [50] use a hybrid CNN to extract features and perform tokenization, which is also used in [51], [52], [53], and [54]. HiT [55] replaces SA blocks in the transformer with convolution operations to improve local spatial and spectral information extraction, named conv-permutator. Although HiT obtains an improved performance, the limitation on capturing long-range dependencies in the spatial and spectral domains remains. The work in [56] and [57] alleviates this problem by embedding convolution operations in the multihead SA (MSA) block of the transformer encoder to improve the attention mechanism for enhancing the local feature extraction capability of conventional ViT. Moreover, recent studies [58], [59] integrate graph-based techniques into transformers to better capture the spatial–spectral information during tokenization, improving the local-to-global feature extraction. GraphGST [59] obtains the physical positional relationship between pixels in generating position encoding by calculating the frequencies of patch labels inside the sliding window.

Although the aforementioned methods achieve great success in the HSI classification, the following issues remain. First, the SA mechanism of the adopted ViT encoder tends to capture global information but is difficult to effectively capture fine-grained local features, especially when the training data is limited. Second, the excessively large amount of parameters, which is caused by the sophisticated neural network architecture design during the tokenization and feature extraction via the transformer encoder, makes them prone to overfitting with limited training data. Although the lightweight transformer-based method [60] has been investigated, the capacity to deal with HSIs of complex structures is limited.

In view of this, we propose a locally enhanced transformer network for HSI classification with a shallow architecture, which consists of a multibranch spatial–spectral tokenization (MSST) module and a dual-branch transformer encoder (DTE) module. Specifically, the proposed MSST module performs shallow-level feature extraction by using three branches with two branches focusing on the spatial and spectral feature extraction and one branch to preserve the original information of HSI. The developed MSST module facilitates the spectral–spatial feature extraction, reducing the information loss during the tokenization process. The dual-branch transformer module, consisting of local and global transformer branches, is designed to fully capture the local and global spatial features of HSI. In the global branch, a regular transformer encoder is adopted to extract the global spatial features of HSI. In the local branch, we explore the local spatial prior information of HSI and develop a locally enhanced transformer encoder based on super-pixel segmentation and graph convolution to extract the local spatial features, which takes into account the local dependencies of pixels in the end-to-end learning process. The extracted local and global spatial features by the two branches are adaptively fused with learnable weights and further fed for HSI classification. To obtain a lightweight neural network, we design MSST and DTE with a shallow architecture, which contains only one layer of CNN and transformer encoder, reducing significantly the number of network parameters. Our experimental results on four benchmark real data sets show that the proposed model yields better performance compared with the state-of-the-art especially when the number of training data is limited. The main contributions of our method are summarized as follows.

1) We propose an MSST module with diverse shallow convolutions and residual connection. This design effectively mitigates information loss during tokenization while maintaining computational efficiency.
2) We develop a DTE module, which consists of a global transformer branch and a locally enhanced transformer branch to capture the local and global data dependencies of HSI, respectively. Moreover, an adaptive fusion strategy with learnable weights is proposed to fuse the local and global features.

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3) To mitigate overfitting with limited training data, we adopt a shallow architecture by using only one layer of CNN and a transformer encoder, significantly reducing the number of parameters. Extensive experimental results demonstrate that our method outperforms the state-of-the-art, especially under limited training data conditions.

The rest of this article is organized as follows. Section II briefly introduces the ViT and GCN for HSI classification. Section III presents the proposed method. Section IV evaluates the classification performance. Section V concludes the article.

II. Preliminaries

A. ViT for HSI Classification

Unlike traditional CNN, which mainly focuses on local feature extraction, ViT [40] is capable of capturing long-range dependencies across both spatial and spectral dimensions in images with transformer encoders. The SA mechanism [41], the core component of the encoder, enables ViT to model relationships between distant pixels by computing attention scores that weigh the importance of each token relative to others in the sequence. This mechanism is effective for HSI, allowing ViT to dynamically focus on inter-band dependencies that are critical for distinguishing subtle spectral variations. We denote the tokens by $\mathbf{X} \in \mathbb{R}^{n\times d}$, where n and d represent the number of tokens and the dimension, respectively. An extra classification token (cls token) and positional embedding in the form of concatenation and addition, respectively, are added to the tokens, and new tokens are denoted by $\mathbf{X}' \in \mathbb{R}^{N\times d}$, where $N = n+1$. Then, the tokens are linearly projected to generate query $\mathbf{Q} \in \mathbb{R}^{N\times d}$, key $\mathbf{K} \in \mathbb{R}^{N\times d}$, and value $\mathbf{V} \in \mathbb{R}^{N\times d}$ by three transformation matrices $\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v \in \mathbb{R}^{d\times d}$. The output $\mathbf{A} \in \mathbb{R}^{N\times d}$ of the one-head SA is obtained by

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V} \in \mathbb{R}^{N\times d} \tag{1}$$

where $\mathbf{Q}\mathbf{K}^T$ calculates the similarity between different tokens and the softmax function normalizes the similarities to a probability distribution. The division by $\sqrt{d}$ prevents large dot product values when d is large. The scaling factor ensures a well-balanced distribution of attention weights, improving convergence stability. The matrix multiplication of the attention weights with $\mathbf{V}$ denotes a linear combination of tokens based on the attention map.

ViT adopts MSA to enhance the feature extraction of data. Unlike single-head SA, which applies a single set of attention weights across the entire feature space, MSA employs multiple independent attention heads, each with its own set of learnable weights. This design allows different attention heads to focus on different aspects of the data. By attending to complementary information at different scales and feature dimensions, MSA significantly enriches the learned feature representations, improving both classification accuracy and model robustness. Specifically, each head independently computes SA as in (1), and the outputs from all heads are concatenated to form a comprehensive representation. Let h represent the number of heads, and MSA can be represented as

$$\text{MSA}(\mathbf{X}') = \text{concat}(\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_h)\mathbf{W} \tag{2}$$

where $\mathbf{W} \in \mathbb{R}^{D\times D}$, $D = h \times d$, and $\mathbf{A}_i$ represents the extracted features of the ith SA. In the implementation of the ViT encoder, layer normalization (LN), residual connection, and multilayer perceptron (MLP) are used. The extracted features of ViT are obtained by

$$\begin{aligned}\mathbf{Y} &= \text{MSA}(\text{LN}(\mathbf{X}')) + \mathbf{X}' \\ \mathbf{Z} &= \text{MLP}(\text{LN}(\mathbf{Y})) + \mathbf{Y}.\end{aligned} \tag{3}$$

The final classification results are obtained from the cls token.

B. Graph Convolutional Network

Traditional convolutional networks operate with fixed kernel shapes. These kernels slide over the regular grid of pixels in an image, performing convolutions to extract features, which works well for data that can be represented on regular grids, like images. However, it has significant limitations when applied to non-Euclidean domains, such as graph-structured data. In these domains, the data points (nodes) do not lie on a regular grid, and the relationships (edges) between them can be complex and irregular [28], [61]. As a result, the fixed kernel shapes of convolutional networks cannot effectively capture the underlying structure of the data.

Graph neural networks (GNNs) are designed to overcome these limitations by operating directly on graph-structured data. A graph is composed of vertices (nodes) representing entities and edges representing the relationships between them. GNNs can learn adaptive kernel parameters based on the specific distribution of nodes and edges, allowing them to perform flexible convolutions on arbitrarily irregular graph structures. This adaptability makes GNNs particularly powerful for modeling complex, non-Euclidean data. Among the various types of GNNs, the most representative GNN is the GCN [62].

GCN extends the concept of convolution from grid-based data to graph-structured data. The core idea is to perform convolution operations on the nodes of a graph, taking into account the graph's topology, which is defined by nodes (representing entities) and edges (representing relationships between entities). In GCN, the convolution operation is adapted to aggregate information from a node's neighbors, thereby capturing the local structure of the graph.

Let graph $G = \{V, E\}$ represent a graph, where V represents the set of all nodes and E represents the edge set. The connections between nodes are represented by an adjacency matrix $\mathbf{S}$, where S_{ij} indicates the presence of an edge between nodes i and j. $\mathbf{S} \in \mathbb{R}^{|V|\times|V|}$ is defined by

$$S_{ij} = \begin{cases} 1, & \text{if nodes } i \text{ and } j \text{ are connected} \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Formally, the lth layer of GCN can be formulated as follows:

$$\mathbf{H}^l = \sigma\left(\hat{\mathbf{S}}\mathbf{H}^{(l-1)}\mathbf{W}^l\right) \tag{5}$$

where $\sigma(\cdot)$ represents a nonlinear activation function, $\mathbf{H}^l$ represents the output of the lth graph convolutional layer,

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$\mathbf{H}^0$ corresponds to the input data $\mathbf{X}$, $\mathbf{W}^l$ denotes the trainable weight matrix of the lth layer, and $\hat{\mathbf{S}}$ represents the symmetric normalized Laplacian regularization matrix $\tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{S}}\tilde{\mathbf{D}}^{-1/2}$. $\tilde{\mathbf{S}}$ is obtained by adding the self-loop to the initial adjacency matrix $\mathbf{S}$. Equation (5) indicates that GCNs aggregate information from all nodes with different weights in a graph, which often facilitates a robust feature extraction for HSI classification.

III. Proposed Method

Here, we introduce our locally enhanced transformer network designed for HSI classification. Fig. 1 shows the overall framework of the proposed network, which comprises two essential modules: an MSST module and a DTE module. The MSST encodes the low-level spatial–spectral information, leading to discriminative tokens. The DTE module combines the global ViT encoder and a neighboring information-guided locally enhanced encoder to fully capture the local and global information of HSI.

A. MSST Module

Transformer has achieved remarkable success in natural language processing. When applying Transformer in image-related tasks, tokenization is an essential step, which can have a significant impact on the subsequent feature extraction by the encoder. The original ViT generates token embeddings by dividing the input image into patches and then performing flatten and linear projection operations on each patch. However, due to the flatten operation, ViT fails to capture the inherent spatial structure of HSI, which is often important for the HSI classification task [63]. In addition, the linear projection used by ViT during the tokenization may further exacerbate the loss of spatial information [44], [55]. While some studies have proposed improved tokenization methods, they still have limitations. For example, the work in [43] employs grouped convolutions to enhance local information extraction but tends to focus on spatial features while neglecting the global spectral information across the entire bands. The works in [42] and [64] use complex pre-processing or convolution operations to maintain the spectral–spatial structure, but these operations increase parameters and computational cost, posing challenges given the limited HSI training data.

To address this problem, we develop an MSST module, which can effectively generate spatial–spectral tokens for subsequent transformer encoders. The MSST module, as shown in Fig. 1(a), consists of three branches with each particularly considering the spectral information, the spatial information, and the raw information of HSI. In order to extract the spatial information, we employ the depth-wise convolution on each channel of the 3-D input patches of HSI. In the second branch, we extract the spectral information of HSI by using point-wise convolution. Note that besides the convolution layers, the batch normalization layer and ReLU activation function are used on the two branches. In order to compensate for the information loss during the spatial and spectral feature extraction within the first two branches, we use a skip connection to propagate the raw information of HSI in the third branch, which is further added with the spatial and spectral features. Let $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$ represent an HSI, where H, W, and B represent the height, width, and the number of bands of HSI, respectively. Let $\mathbf{X}_i \in \mathbb{R}^{p\times p\times B}$ be an HSI patch of the ith pixel, where p is the size of the square window. The MSST module is represented as follows:

$$F(\mathbf{X}_i) = F_{\text{spa}}(\mathbf{X}_i) + F_{\text{spe}}(\mathbf{X}_i) + \mathbf{X}_i \tag{6}$$

where F_{spa} and F_{spe} denote the spatial and spectral feature extraction in the two branches, respectively. Note that proper settings on padding, stride, and channels are needed to ensure the same output dimensions across the three branches.

After the fusion of the features from the three branches in (6), a 1×1 convolution is used to reduce the dimension of $F(\mathbf{X}_i)$, resulting in enhanced tokens $\mathbf{X}^i_{\text{seq}} \in \mathbb{R}^{p^2\times dim}$, which will be fed into the subsequent DTE module. Our developed MSST module is simple but effective. It fully considers the spatial, spectral, and raw information of HSI, facilitating to obtaining of more discriminative tokens for the feature extraction of the subsequent transformer encoder as evidenced by our experiments in Table X. In addition, we adopt a shallow architecture with each branch consisting of one convolution layer, which indicates that our tokenization module contains fewer parameters and leads to an overall low computational cost, as shown in Table XII.

B. DTE Module

The SA in the conventional MSA, as depicted in (1), aggregates information by a linear combination of all the tokens, which allows for effective extraction of the global information of the data. However, the conventional MSA in (2) fails to exploit the local spatial information of HSI, which means that the local spatial dependencies between tokens are not taken into account. HSIs often show large within-class spectral variance due to the effect of noise and spectral variabilities. Extracting features globally, as in the conventional MSA, can lead to the information aggregation from irrelevant tokens and thus degrade the discriminability of the learned features. Such performance degradation is more pronounced when only a few training data are available [40], [65]. To address this problem, we propose a DTE module to enhance the local spatial information extraction ability. As shown in the right of Fig. 1(a), the developed encoder module consists of a global branch and a local branch, where the global branch adopts the ViT encoder with the conventional MSA to extract the global information of HSI and the local branch focuses on the local information extraction of HSI by using a locally enhanced transformer encoder.

1) Local Transformer Encoder: In order to encode the local information of HSI, we propose an improved MSA (IMSA) module by taking into account the local spatial prior information of HSI. We exploit the local dependencies of pixels within a graph based on super-pixel segmentation and then embed the constructed graph in the IMSA by using a linear graph convolution.

We first use the super-pixel segmentation technique to generate a local graph where the pixels are viewed as vertices. Specifically, we adopt the principal component analysis (PCA)

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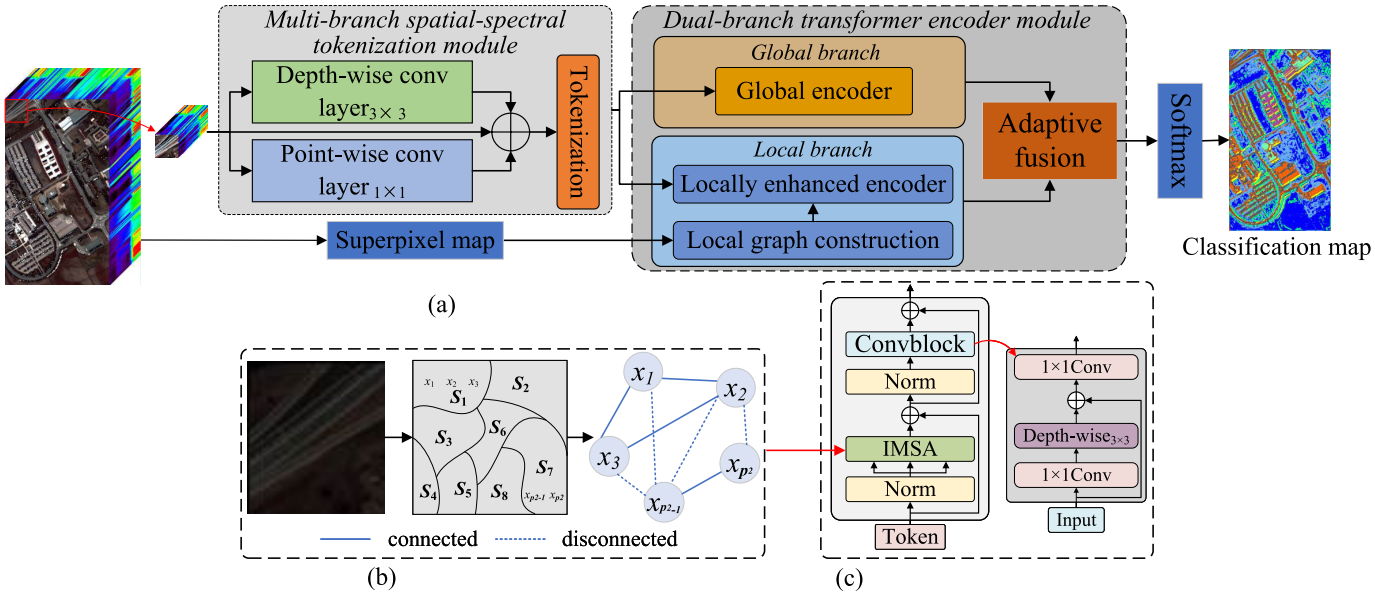


Fig. 1. Overall framework of the proposed locally enhanced transformer network for HSI classification. (a) Model consists of an MSST module and a DTE. The tokenization module extracts spectral and spatial features through depth-wise and point-wise convolution layers. The DTE consists of a global transformer encoder and a locally enhanced transformer encoder, which focus on capturing the global and local structures of HSI, respectively. (b) Local graph is constructed based on superpixel segmentation, providing local prior information for the local branch. (c) We develop an IMSA with local graph convolution and a Convblock to improve the spatial–spectral feature extraction in the locally enhanced encoder.

to reduce the dimension of HSI to one and then apply the entropy rate super-pixel segmentation method [66] in the first PC of HSI to generate super-pixels $\{\mathcal{S}_i\}_{i=1}^{K}$, where K is the number of super-pixels. As the pixels within a super-pixel often belong to the same class, we make use of such local spatial prior information to construct a local graph $\mathbf{M} \in \mathbb{R}^{p^2 \times p^2}$ for each input patch $\mathbf{X} \in \mathbb{R}^{p \times p \times B}$, where $p \times p$ is the window size for the central pixel. Note that we remove the explicit index of the input patch for simplicity. As pixels belonging to the same super-pixels are often of high similarity, we build the similarity matrix $\mathbf{M} \in \mathbb{R}^{p^2 \times p^2}$ of the input patch as follows:

$$\mathbf{M}_{ij} = \begin{cases} e^{-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}}, & \mathbf{x}_i, \mathbf{x}_j \in \mathcal{S}_t \\ 0, & \text{otherwise} \end{cases}$$
$$\sigma^2 = \frac{1}{p^4} \sum_i \sum_j \left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2 \tag{7}$$

where $\mathbf{x}_i$ and $\mathbf{x}_j$ are the ith and jth pixels of an HSI patch, and σ is the mean value of the pairwise distance of pixels within a patch. Equation (7) indicates that if two pixels in a patch belong to the same super-pixel, they are connected in the graph, otherwise, their similarity is set to zero. The local graph construction is shown in Fig. 1(b).

After the construction of the local graph in (7), we improve the local feature extraction of the commonly used MSA module by using graph convolution. Graph convolution aggregates information from all the nodes according to a predefined graph, where a larger weight makes a higher contribution. As the similarity matrix $\mathbf{M}$ encodes the local dependencies of pixels, we embed it in the SA model to improve the local information extraction. Our proposed SA module, which considers the local prior information of HSI, is formulated as follows:

$$\hat{\mathbf{A}} = \left(\mathbf{D}^{-1/2} \widetilde{\mathbf{M}} \mathbf{D}^{-1/2}\right) \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right) \mathbf{V} \tag{8}$$

where $\widetilde{\mathbf{M}} = \mathbf{I} + \mathbf{M}$ is a self-looping adjacency matrix, and $\mathbf{I}$ is an identity matrix and $\mathbf{D}$ is a diagonal degree matrix of $\widetilde{\mathbf{M}}$ with $\mathbf{D}_{ii} = \sum_j \widetilde{\mathbf{M}}_{ij}$. $(\mathbf{D}^{-1/2} \widetilde{\mathbf{M}} \mathbf{D}^{-1/2})$ is a normalized similarity matrix and softmax$(\mathbf{Q}\mathbf{K}^T/\sqrt{d})\mathbf{V}$ is the conventional SA module. The graph convolution in (8) is a simplified version of (5) by removing the nonlinear activation function. With the linear graph convolution, the obtained features of a data point by (8) aggregate more information from its neighbors by using larger weights, as defined by (7), leading to more robust features to noise and spectral variability.

The improved SA in (8) can be easily extended to MSA, resulting in an IMSA. Moreover, we develop a convblock, which consists of point-wise and depth-wise convolutions, as shown in Fig. 1(c), to further improve the spatial–spectral feature extraction of the local transformer encoder. Unlike the traditional MLP block with two linear layers and an additional activation function layer, the point-wise and depth-wise convolutions focus on the feature extraction in the channel and spatial dimensions, respectively. A residual connection is introduced to avoid information loss during the feature transformation. To this end, the proposed local branch is formulated by

$$\begin{aligned} \text{IMSA}\left(\mathbf{X}_{\text{seq}}^i\right) &= \text{concat}\left(\hat{\mathbf{A}}_1, \hat{\mathbf{A}}_2, \ldots, \hat{\mathbf{A}}_h\right)\mathbf{W} \\ \mathbf{Z} &= \text{IMSA}\left(\text{LN}\left(\mathbf{X}_{\text{seq}}^i\right)\right) + \mathbf{X}_{\text{seq}}^i \\ \mathbf{F}_{\text{local}} &= \text{Convblock}(\text{LN}(\mathbf{Z})) + \mathbf{Z} \end{aligned} \tag{9}$$

where $\hat{\mathbf{A}}_i$ represents the features of the ith improved SA head and $\mathbf{F}_{\text{local}}$ represents the output features of the local branch.

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Note that to fit with the expanded dimension of the input patch because of the classification token, we expand $\mathbf{M}$ by setting the diagonal elements to one and the other expanded elements to zeros.

2) Global Encoder: The local branch effectively encodes the local information of HSI by the improved SA module. In order to capture the long-range spatial dependencies of tokens, we employ the conventional transformer encoder in the global feature extraction branch, which is in parallel with the local feature extraction branch. The input of the global transformer encoder is the same as the local branch, which is $\mathbf{X}^i_{\text{seq}}$ obtained in Section III-A. With the conventional multihead SA, the global transformer encoder captures the long-range spatial dependencies as follows:

$$\begin{aligned} \text{MSA}\left(\mathbf{X}^i_{\text{seq}}\right) &= \text{concat}(\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_h)\mathbf{W} \\ \mathbf{Y} &= \text{MSA}\left(\text{LN}\left(\mathbf{X}^i_{\text{seq}}\right)\right) + \mathbf{X}^i_{\text{seq}} \\ \mathbf{F}_{\text{global}} &= \text{MLP}(\text{LN}(\mathbf{Y})) + \mathbf{Y} \end{aligned} \tag{10}$$

where $\mathbf{A}_i$ is the output of the ith SA module and $\mathbf{F}_{\text{global}}$ is the extracted features of the global transformer encoder.

Most existing transformer-based methods of HSI cascade multiple transformer encoders to extract multiscale and deeper features of HSI. Although they obtain improved performance, the amount of parameters in the networks is significantly increased, which often demands more training data to avoid over-fitting. In our work, we set the number of transformer encoders to one in the local and global branches, which further reduces the amount of parameters in the network.

C. Adaptive Fusion

The two branches encode the spatial dependencies of pixels at different ranges, where the local branch captures the fine details and local contextual information, and the global branch extracts features at a broader view. In order to leverage the complementary information from the two branches, we fuse the information with adaptive weights as follows:

$$\mathbf{F} = w_1\mathbf{F}_{\text{local}} + w_2\mathbf{F}_{\text{global}} \tag{11}$$

where w_1 and w_2 are two learnable weights. These weights are initialized using a normal distribution and trained through back-propagation. Specifically, w_1 and w_2 are initialized as learnable parameters, which are randomly initialized at the start of training. During the training process, these weights are updated based on the gradients computed from the loss function, allowing the model to adaptively learn the optimal contribution of local and global features. The final classification results are obtained from the classification token via the classifier consisting of a linear layer and a softmax function. A pseudocode of our methods is provided in Algorithm 1.

IV. Experiments and Analysis

We conduct experiments on four benchmark HSI datasets, including *Pavia University* (PaviaU), *University of Houston*, *WHU-Hi-LongKou*, and a new challenging HSI dataset, *ShanDongFeiCheng* (SDFC) dataset, to evaluate the performance of our proposed method.

Algorithm 1 Pseudocode of Our Method

Input: HSI data $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$ and ground truth $\mathbf{Y} \in \mathbb{R}^{H\times W}$
Output: Classification accuracy and classification map.
Generate training and testing data sets.
Obtain superpixel segmentation results via [66].
for *epoches* **do**
    Feed $\mathbf{X}_i$ in the training data set to MSST and obtain $F(\mathbf{X}_i) = F_{spa}(\mathbf{X}_i) + F_{spe}(\mathbf{X}_i) + \mathbf{X}_i$
    Use 1×1 conv to reduce the dimension of $F(\mathbf{X}_i)$
    Flatten $F(\mathbf{X}_i)$ in spatial dimension as feature tokens $\mathbf{X}^i_{seq}$
    Obtain F_{global} by passing $\mathbf{X}^i_{seq}$ through the global encoder
    Obtain F_{local} by feeding $\mathbf{X}^i_{seq}$ into the locally enhanced encoder.
    Perform feature fusion via Eq. (11)
    Input the cls token to the last linear layer and softmax function to identify the labels.
end for
Save the parameters of the model and obtain the predicted labels of testing data. Calculate classification accuracy and obtain the visual classification map.

TABLE I
BASIC INFORMATION OF THE FOUR HSI DATASETS

Dataset	PaviaU	Houston2013	LongKou	SDFC
Spatial size	610×340	349×1905	550×400	1201×601
Bands	103	144	270	63
Labelled pixels	42776	15029	204542	652743
Classes	9	15	9	20
Spectral range	430–860 nm	$380-1050$ nm	$400-1000$ nm	$400-1000$ nm
Sensor	ROSIS	CASI	Nano-HP	HAHS
Spatial resolution	1.3 m	2.5 m	0.463 m	0.5 m

TABLE II
NUMBER OF TRAINING AND TESTING DATA IN PAVIAU

Class ID	Class name	Train	Test
1	Asphalt	20	6611
2	Meadows	20	18629
3	Gravel	20	2079
4	Trees	20	3044
5	Painted metal sheets	20	1325
6	Bare soil	20	5009
7	Bitumen	20	1310
8	Self-blocking bricks	20	3662
9	Shadows	20	927
	Total	180	42596

A. Hyperspectral Datasets

Table I shows the basic information of the four public datasets.

1) Pavia University: This HSI was collected by the reflective optics system imaging spectrometer (ROSIS) sensor in 2002 and recorded scenes over the University of Pavia, Italy. The data size is $610 \times 340 \times 103$. The false color image and ground truth are shown in Fig. 2(a) and (b), which include nine classes. The data split for the training set and testing set is shown in Table II.

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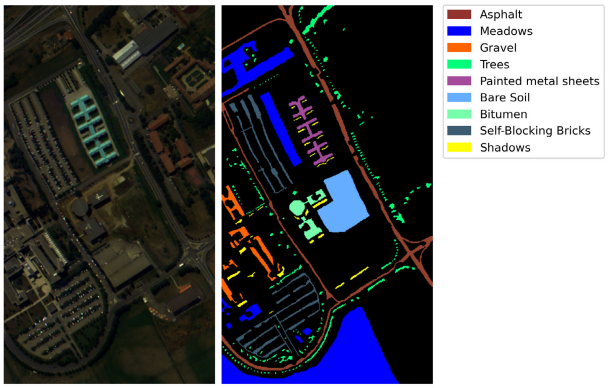


Fig. 2. *PaviaU University* dataset. (a) False color image. (b) Ground truth.

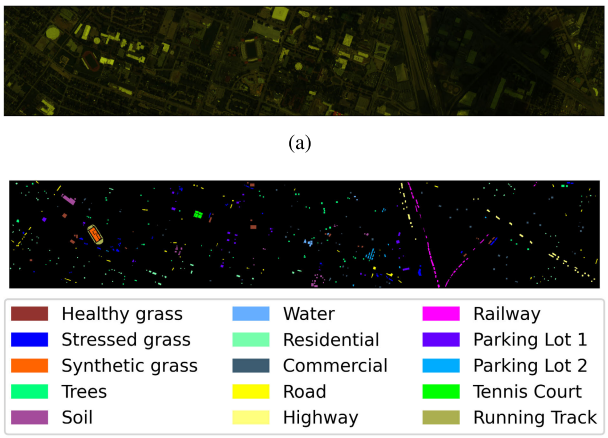


Fig. 3. *University of Houston* dataset. (a) False color image. (b) Ground truth.

2) University of Houston (Houston2013): The HSI data was acquired by the ITRES-CASI 1500 sensor over the University of Houston campus and the neighboring urban area, and it was released in the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest. Houston2013 consists of 349 × 1905 pixels and has 144 spectral bands ranging from 380 to 1050 nm. The false color image and the ground truth are shown in Fig. 3(a) and (b). The data split for the training set and testing set for each category is shown in Table III.

3) WHU-Hi-LongKou (LongKou): The hyperspectral data was acquired in 2018 in LongKou Town, Hubei Province, China, using an 8-mm focal length Headwall Nano-Hyperspec imaging sensor mounted on a UAV platform [67]. The size of the imagery is 550 × 400 with 270 bands (from 400 to 1000 nm). The false color image and the ground truth are shown in Fig. 4(a) and (b). We show the amount of data for each class in the training set and testing set in Table IV.

4) ShanDongFeiCheng: The SDFC data [68] was published in 2023 and was acquired in 2018 using the High-spectral Aerial Hyperspectral Sensor (HAHS) over the FeiCheng area in Shandong Province, China. The dataset consists of 1201 × 601 pixels with 63 spectral bands ranging from 400 to 1000 nm, with a large-scale dataset size and high spatial resolution. It contains 20 categories of annotated pixels, with 90% of the entire image annotated. The false color image and the ground truth are shown in Fig. 5(a) and (b). We show the amount of data for each class in the training set and testing set in Table V.

TABLE III
NUMBER OF TRAINING AND TESTING DATA IN HOUSTON2013

Class ID	Class name	Train	Test
1	Grass healthy	20	1231
2	Grass stressed	20	1234
3	Grass synthetic	20	677
4	Tree	20	1224
5	Soil	20	1222
6	Water	20	305
7	Residential	20	1248
8	Commercial	20	1224
9	Road	20	1232
10	Highway	20	1207
11	Railway	20	1215
12	Parking lot 1	20	1213
13	Parking lot 2	20	449
14	Tennis court	20	408
15	Running track	20	640
	Total	300	14729

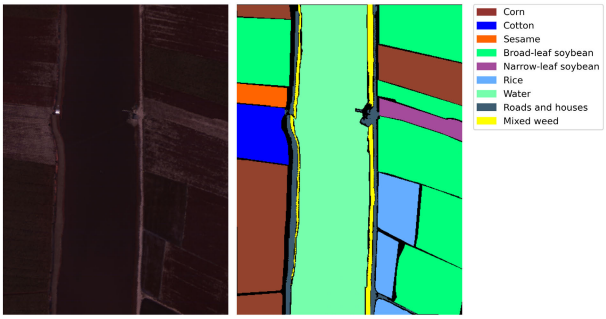


Fig. 4. *WHU-Hi-LongKou* dataset. (a) False color image. (b) Ground truth.

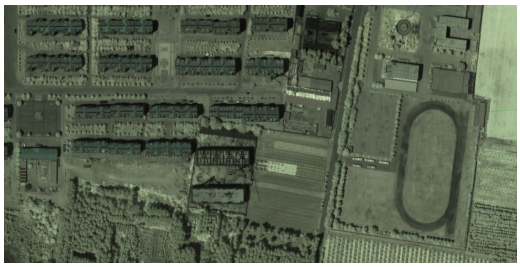
TABLE IV
NUMBER OF TRAINING AND TESTING DATA IN LONGKOU

Class ID	Class name	Train	Test
1	Corn	20	34491
2	Cotton	20	8354
3	Sesame	20	3011
4	Broad-leaf soybean	20	63192
5	Narrow-leaf soybean	20	4131
6	Rice	20	11834
7	Water	20	67026
8	Roads and houses	20	7104
9	Mixed weed	20	5209
	Total	180	204362

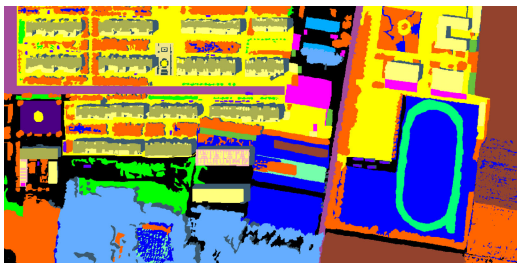
B. Experiment Setting

1) Compared Methods: To validate the performance of our proposed model, we compare it with representative state-of-the-art deep HSI classification algorithms, including two CNN-based methods: HybridSN [32], DFFN [69], and six transformer-based methods: Spectralformer [48],

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(a)



(b)

Fig. 5. *SDFC* dataset. (a) False color image. (b) Ground truth.

TABLE V
NUMBER OF TRAINING AND TESTING DATA IN SDFC

Class ID	Class name	Train	Test
1	Crops 1	20	56644
2	Bare Soil	20	97564
3	Shrubs	20	115878
4	Playground Track	20	13826
5	Road	20	25178
6	Trees	20	64145
7	Crops 2	20	4810
8	Shadow	20	32929
9	Concrete Floor	20	114604
10	Concrete Roof	20	38115
11	Caigang Watt Roof	20	10549
12	Cars	20	1905
13	Asphalt Roof	20	2916
14	Grass	20	18726
15	Glazed Tile Roof	20	38214
16	Clay Tile Roof	20	3897
17	Special Material	20	3031
18	Piazza Area	20	6463
19	White Tile Floor	20	2190
20	Black Tile Floor	20	759
	Total	400	652343

HiT [55], SSFTT [50], GAHT [43], Morphformer [45], and GraphGST [59]. HybridSN is a hybrid CNN model based on 3D-CNN and 2D-CNN, and DFFN employs a deep CNN-based architecture that introduces residual learning to optimize deep network training and extract features at different depths.

2) Evaluation Metrics: The quantitative metrics to measure the classification performance are each class accuracy (ECA), overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ). OA refers to the ratio of the number of correctly classified pixels to the number of pixels. AA is the average of the classification accuracy for each category, reflecting the ability of the classifier to recognize different categories, and κ is used to quantitatively assess the consistency of the predicted results with the true label. The higher the value of these metrics, the better the classification performance. For a dataset with C category samples, let the number of samples in its cth category be N_c ($1 \le c \le C$) and the number of correctly categorized samples in that category is M_c ($M_c \le N_c$), every metric is calculated as follows:

$$\mathrm{OA} = \frac{\sum_c M_c}{\sum_c N_c} \tag{12}$$

$$\mathrm{AA} = \frac{1}{C}\sum_c \mathrm{ECA}_c \tag{13}$$

$$\kappa = \frac{\mathrm{OA} - P_e}{1 - P_e} \tag{14}$$

where $\mathrm{ECA}_c = (M_c/N_c)$ denotes the classification accuracy of the cth class and $P_e = (\sum_c M_c N_c/(\sum_c N_c)^2)$.

3) Parameter Setting: The proposed model is implemented in Python. We run all the experiments on a server with an NVIDIA RTX 4090 GPU. In the experiments, 20 samples per class are used for training and the rest are used for testing. Detailed category information of the four datasets is shown in Tables II–V. The Adam optimizer is used with an initial learning rate of 0.0005, which is decayed by 0.0001. The batch size is set to 64. All experiments are run ten times, and the average performance is reported.

C. Performance Comparison

1) Quantitative Evaluation: We report the means and standard deviations of the classification accuracies of the different methods on the PaviaU, Houston2013, LongKou, and SDFC datasets in Tables VI–IX, respectively. The best results are marked in bold. The results show that our proposed method achieves the best results on PaviaU, LongKou, and SDFC in terms of OA, AA, and κ. For the CNN-based methods, both HybridSN and DFFN obtain satisfactory classification accuracies. It is observed that DFFN outperforms HybridSN on the four datasets, which can be mainly attributed to the larger receptive field by using a deeper CNN architecture. Our model outperforms DFFN by 2.71% OA improvement on PaviaU, 3.29% OA improvement on Houston2013, and 3.41% OA improvement on LongKou, which mainly benefits from the improved ability to capture long-range contextual information of HSI with a transformer. We also notice that not all transformer-based methods perform better than the CNN-based methods. For instance, HiT and Spectralformer often yield worse accuracies than HybridSN and DFFN. This can be attributed to the fact that in HiT, the SA blocks are replaced by convolutional operations, which limit the ability to extract long-range spatial information. Furthermore, Spectralformer

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TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON PAVIAU DATASET

Class	HybridSN	DFFN	HiT	Spectralformer	SSFTT	GAHT	Morphformer	GraphGST	Proposed
1	83.77 ± 5.82	91.38 ± 0.42	52.04 ± 12.19	71.55 ± 6.62	89.34 ± 3.58	92.36 ± 2.69	89.75 ± 3.03	88.71 ± 0.97	$\mathbf{94.19 \pm 1.79}$
2	93.53 ± 4.97	94.01 ± 2.44	69.48 ± 12.08	82.02 ± 4.34	94.42 ± 0.90	92.51 ± 1.19	88.90 ± 6.83	94.68 ± 0.91	$\mathbf{96.25 \pm 1.41}$
3	94.62 ± 4.13	93.63 ± 2.55	78.32 ± 5.25	75.35 ± 7.33	$\mathbf{94.96 \pm 0.94}$	80.78 ± 9.30	89.92 ± 7.46	87.67 ± 1.04	93.02 ± 4.21
4	83.37 ± 6.65	80.00 ± 2.52	93.55 ± 1.30	95.29 ± 1.72	95.96 ± 0.98	$\mathbf{98.10 \pm 0.64}$	95.82 ± 0.80	95.36 ± 1.32	92.28 ± 2.40
5	99.43 ± 0.31	89.81 ± 4.67	99.52 ± 0.19	$\mathbf{100.0 \pm 0.00}$	99.70 ± 0.10	99.98 ± 0.03	99.72 ± 0.17	$\mathbf{100.0 \pm 0.00}$	99.57 ± 0.27
6	94.36 ± 3.03	99.04 ± 0.59	79.47 ± 9.88	73.38 ± 7.68	$\mathbf{99.09 \pm 0.97}$	90.84 ± 12.03	91.39 ± 5.18	89.46 ± 1.74	98.78 ± 0.72
7	99.79 ± 0.31	98.01 ± 1.01	78.19 ± 4.86	75.22 ± 5.20	$\mathbf{99.98 \pm 0.05}$	93.40 ± 2.37	96.82 ± 1.00	98.50 ± 0.28	99.10 ± 0.69
8	66.73 ± 13.59	96.14 ± 0.69	72.63 ± 7.03	67.16 ± 10.18	78.12 ± 4.89	89.63 ± 5.42	91.80 ± 4.22	93.56 ± 0.65	$\mathbf{96.15 \pm 1.69}$
9	83.52 ± 6.25	87.50 ± 4.39	87.81 ± 1.82	96.88 ± 1.41	97.04 ± 1.12	$\mathbf{99.76 \pm 0.20}$	97.02 ± 1.04	99.49 ± 0.10	93.87 ± 3.00
OA	89.30 ± 3.13	93.21 ± 0.92	71.97 ± 4.99	79.40 ± 1.36	93.31 ± 1.08	92.28 ± 1.92	90.87 ± 3.25	93.14 ± 0.36	$\mathbf{95.92 \pm 0.88}$
AA	88.79 ± 1.88	92.17 ± 0.58	79.00 ± 2.17	81.87 ± 1.29	94.29 ± 0.98	93.04 ± 2.04	93.46 ± 1.52	94.16 ± 0.23	$\mathbf{95.91 \pm 0.91}$
κ	86.00 ± 3.92	91.08 ± 1.16	64.94 ± 5.42	73.40 ± 1.61	91.23 ± 1.40	89.87 ± 2.57	88.17 ± 4.06	90.98 ± 0.46	$\mathbf{94.62 \pm 1.14}$

TABLE VII
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON HOUSTON2013 DATASET

Class	HybridSN	DFFN	HiT	Spectralformer	SSFTT	GAHT	Morphformer	GraphGST	Proposed
1	89.76 ± 6.56	92.30 ± 1.38	82.49 ± 6.21	93.49 ± 2.60	94.96 ± 2.69	94.32 ± 4.43	95.61 ± 1.73	92.74 ± 3.99	$\mathbf{95.71 \pm 1.78}$
2	91.77 ± 1.81	84.91 ± 3.08	71.10 ± 12.53	91.64 ± 3.75	93.30 ± 1.87	90.46 ± 3.53	93.36 ± 4.07	$\mathbf{95.23 \pm 1.32}$	92.19 ± 1.67
3	98.54 ± 0.97	98.95 ± 0.33	91.23 ± 6.44	95.07 ± 1.81	$\mathbf{99.38 \pm 0.62}$	99.01 ± 0.84	99.29 ± 0.46	92.11 ± 0.22	99.32 ± 0.07
4	78.26 ± 7.01	83.13 ± 3.20	70.56 ± 8.56	84.57 ± 6.48	$\mathbf{95.27 \pm 1.43}$	88.33 ± 2.58	91.39 ± 1.84	89.09 ± 1.43	93.95 ± 1.12
5	94.53 ± 4.78	95.05 ± 5.29	92.13 ± 3.84	96.42 ± 1.87	96.08 ± 3.34	96.48 ± 1.97	91.46 ± 8.38	$\mathbf{100.0 \pm 0.00}$	98.12 ± 1.52
6	99.48 ± 0.83	99.31 ± 0.77	85.08 ± 3.60	93.97 ± 3.88	$\mathbf{100.0 \pm 0.00}$	96.52 ± 2.04	97.08 ± 1.16	$\mathbf{100.0 \pm 0.00}$	98.98 ± 0.50
7	77.30 ± 8.85	65.22 ± 3.81	66.95 ± 9.35	71.97 ± 5.08	77.50 ± 8.96	76.83 ± 4.62	$\mathbf{83.71 \pm 2.15}$	79.46 ± 5.41	79.80 ± 1.73
8	63.31 ± 5.58	63.49 ± 4.01	38.80 ± 7.77	66.18 ± 3.41	63.42 ± 4.80	66.79 ± 2.39	66.40 ± 4.51	62.79 ± 1.71	$\mathbf{69.79 \pm 2.13}$
9	63.29 ± 7.71	78.98 ± 3.37	68.78 ± 13.30	60.82 ± 10.95	79.84 ± 4.38	79.26 ± 5.62	81.47 ± 7.13	$\mathbf{85.92 \pm 8.49}$	83.51 ± 2.80
10	86.64 ± 4.63	$\mathbf{96.13 \pm 3.96}$	65.78 ± 9.24	81.92 ± 4.68	92.44 ± 3.95	88.85 ± 13.34	88.91 ± 11.38	93.89 ± 2.59	93.91 ± 3.97
11	90.82 ± 3.96	93.07 ± 3.29	48.82 ± 19.06	65.21 ± 6.48	93.70 ± 3.21	77.29 ± 5.63	86.32 ± 3.08	$\mathbf{96.21 \pm 0.84}$	85.35 ± 3.61
12	92.41 ± 4.03	$\mathbf{92.69 \pm 1.05}$	41.57 ± 21.87	66.74 ± 8.14	92.69 ± 1.76	85.42 ± 5.92	83.78 ± 6.55	90.34 ± 14.38	90.88 ± 1.58
13	87.28 ± 3.64	88.86 ± 1.74	74.61 ± 21.01	68.55 ± 11.01	84.92 ± 1.95	86.84 ± 6.27	84.54 ± 4.99	$\mathbf{94.86 \pm 0.39}$	87.17 ± 2.52
14	$\mathbf{100.0 \pm 0.00}$	$\mathbf{100.0 \pm 0.00}$	97.43 ± 1.95	96.86 ± 2.32	$\mathbf{100.0 \pm 0.00}$	98.92 ± 0.61	99.36 ± 1.10	$\mathbf{100.0 \pm 0.00}$	99.61 ± 0.41
15	98.84 ± 1.33	95.89 ± 1.62	91.56 ± 3.22	98.64 ± 1.34	$\mathbf{99.88 \pm 0.15}$	99.50 ± 0.88	98.45 ± 2.21	99.17 ± 0.12	98.72 ± 0.91
OA	85.16 ± 1.51	86.47 ± 0.94	68.75 ± 2.20	80.17 ± 3.13	89.43 ± 1.72	86.45 ± 2.07	87.91 ± 1.82	$\mathbf{89.91 \pm 1.65}$	89.76 ± 0.49
AA	87.48 ± 1.19	88.53 ± 0.82	72.46 ± 2.63	82.13 ± 3.04	90.89 ± 1.41	88.32 ± 1.81	89.41 ± 1.44	$\mathbf{91.45 \pm 1.34}$	91.13 ± 0.45
κ	83.96 ± 1.62	85.38 ± 1.02	66.27 ± 2.40	78.57 ± 3.39	88.57 ± 1.86	85.36 ± 2.22	86.93 ± 1.96	$\mathbf{89.10 \pm 1.79}$	88.93 ± 0.53

employs a simple tokenization process that neglects the spatial information inherent in HSI. In contrast, SSFTT, GAHT, Morphformer, and GraphGST improve the tokenization and attention mechanisms, leading to improved performance over HiT and Spectralformer.

Compared with the transformer-based methods, our model significantly improves the performance in terms of OA, AA, and κ. On the PaviaU dataset, our method outperforms the second-best method SSFTT by 2.61% in OA, 1.62% in AA, and 3.39% in κ. Compared with GraphGST, which incorporates graph structure information of data in the model, our method also obtains a better performance. Similar results can be observed on the LongKou dataset. On the Houston2013 dataset, our method performs comparably to the GraphGST. On the new challenging SDFC dataset, our method performs the best with OA improvement by 0.78% compared with the second-best method DFFN. Our method also surpasses transformer-based methods across all three metrics. For instance, it surpasses GraphGST by 5.08%, 5.79%, and 5.35% on OA, AA, and κ, respectively. In general, the improvement of our method can be mainly attributed to the MSST, DTE and adaptive fusion strategy. The MSST enhances the processing of spatial and spectral information of HSI during tokenization while reducing information loss. DTE leverages local dependencies prior information, which enhances the learning of local context features. The adaptive fusion strategy allows the model to dynamically adjust the importance of the two branches, further boosting performance.

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TABLE VIII
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON LONGKOU DATASET

Class	HybridSN	DFFN	HiT	Spectralformer	SSFTT	GAHT	Morphformer	GraphGST	Proposed
1	95.21 ± 3.47	97.70 ± 1.95	98.32 ± 0.97	97.31 ± 0.99	99.85 ± 0.10	99.77 ± 0.14	98.16 ± 1.01	99.72 ± 0.10	$\mathbf{99.96 \pm 0.09}$
2	93.65 ± 5.65	95.67 ± 1.83	75.41 ± 17.80	91.75 ± 2.21	$\mathbf{98.41 \pm 0.61}$	97.50 ± 0.84	95.55 ± 1.60	95.92 ± 0.54	97.87 ± 0.88
3	$\mathbf{99.73 \pm 0.58}$	98.80 ± 0.82	90.35 ± 6.51	90.82 ± 2.77	99.48 ± 0.89	91.81 ± 1.68	94.19 ± 4.21	98.22 ± 0.16	98.60 ± 1.45
4	86.46 ± 3.63	90.90 ± 1.64	83.70 ± 2.81	82.84 ± 1.68	93.52 ± 1.31	90.73 ± 1.46	93.58 ± 1.59	88.29 ± 0.50	$\mathbf{96.48 \pm 1.53}$
5	93.31 ± 4.89	$\mathbf{99.87 \pm 0.22}$	86.72 ± 4.92	86.13 ± 1.68	98.03 ± 0.85	92.05 ± 4.98	87.87 ± 10.85	99.26 ± 0.13	96.12 ± 2.62
6	98.53 ± 1.88	94.34 ± 2.02	94.02 ± 9.43	98.09 ± 0.81	99.64 ± 0.24	99.61 ± 0.28	98.61 ± 0.69	$\mathbf{99.65 \pm 0.15}$	99.46 ± 0.49
7	96.41 ± 1.52	97.57 ± 0.43	99.64 ± 0.32	99.35 ± 0.26	97.58 ± 0.48	99.48 ± 0.10	99.15 ± 0.44	$\mathbf{99.97 \pm 0.00}$	98.54 ± 0.26
8	84.66 ± 6.14	76.91 ± 2.25	92.43 ± 2.71	94.91 ± 1.89	95.53 ± 1.32	$\mathbf{97.21 \pm 0.79}$	94.96 ± 1.32	91.84 ± 0.79	95.86 ± 2.23
9	88.86 ± 2.90	90.76 ± 2.09	88.77 ± 1.75	85.71 ± 4.27	87.96 ± 1.89	89.18 ± 2.35	83.95 ± 3.88	$\mathbf{94.83 \pm 0.62}$	91.65 ± 1.28
OA	92.52 ± 1.42	94.44 ± 0.56	92.24 ± 1.86	92.62 ± 0.67	96.58 ± 0.43	96.14 ± 0.49	96.25 ± 0.45	95.68 ± 0.16	$\mathbf{97.85 \pm 0.49}$
AA	92.98 ± 1.25	93.61 ± 0.56	89.93 ± 3.37	91.88 ± 0.68	96.67 ± 0.25	95.26 ± 0.85	94.00 ± 1.00	96.41 ± 0.06	$\mathbf{97.17 \pm 0.34}$
κ	90.29 ± 1.83	92.78 ± 0.72	89.97 ± 2.36	90.48 ± 0.85	95.55 ± 0.56	94.88 ± 0.63	95.10 ± 0.58	94.39 ± 0.20	$\mathbf{97.19 \pm 0.64}$

TABLE IX
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON SDFC DATASET

Class	HybridSN	DFFN	HiT	Spectralformer	SSFTT	GAHT	Morphformer	GraphGST	Proposed
1	65.10 ± 6.00	85.68 ± 5.11	84.68 ± 4.43	62.40 ± 9.37	61.52 ± 10.34	78.13 ± 3.31	84.15 ± 1.97	$\mathbf{89.12 \pm 2.39}$	85.57 ± 3.14
2	50.91 ± 8.63	60.89 ± 3.04	16.85 ± 26.32	45.88 ± 8.62	61.25 ± 7.73	53.74 ± 19.88	$\mathbf{69.65 \pm 4.18}$	48.70 ± 6.73	68.49 ± 2.85
3	23.27 ± 4.53	42.63 ± 4.46	9.85 ± 6.44	19.23 ± 2.67	44.37 ± 8.69	33.98 ± 10.50	34.13 ± 6.07	$\mathbf{45.53 \pm 3.48}$	41.80 ± 3.33
4	65.57 ± 8.45	$\mathbf{96.70 \pm 2.07}$	37.97 ± 18.67	49.19 ± 12.93	72.87 ± 8.26	69.41 ± 6.24	90.39 ± 4.42	73.68 ± 3.55	86.91 ± 3.61
5	49.97 ± 12.21	$\mathbf{85.73 \pm 4.92}$	42.38 ± 20.15	50.35 ± 6.23	64.63 ± 9.15	73.67 ± 10.92	75.92 ± 6.38	53.27 ± 9.90	81.28 ± 2.33
6	69.44 ± 5.83	$\mathbf{85.48 \pm 2.03}$	39.28 ± 11.77	42.59 ± 8.57	76.28 ± 6.40	72.83 ± 9.32	79.66 ± 3.37	77.68 ± 2.04	85.46 ± 1.53
7	70.90 ± 5.93	88.28 ± 4.15	33.16 ± 20.57	37.82 ± 6.50	72.62 ± 13.93	78.01 ± 8.62	84.01 ± 5.72	88.99 ± 4.01	$\mathbf{89.16 \pm 2.10}$
8	38.56 ± 6.45	52.78 ± 3.25	28.59 ± 9.08	44.42 ± 7.59	37.49 ± 11.14	52.13 ± 5.26	54.01 ± 4.78	$\mathbf{54.02 \pm 2.62}$	42.84 ± 2.41
9	32.13 ± 5.01	32.68 ± 5.05	25.72 ± 22.74	21.18 ± 3.00	33.84 ± 9.59	36.29 ± 10.36	38.64 ± 7.05	38.42 ± 5.38	$\mathbf{41.91 \pm 3.62}$
10	28.68 ± 6.47	$\mathbf{45.98 \pm 4.71}$	9.15 ± 6.08	19.97 ± 3.79	26.80 ± 11.74	32.12 ± 7.06	34.10 ± 9.35	23.06 ± 6.38	40.01 ± 2.64
11	$\mathbf{84.29 \pm 4.77}$	76.55 ± 4.12	16.23 ± 14.43	35.63 ± 9.29	76.88 ± 4.48	60.65 ± 8.13	57.71 ± 7.23	73.42 ± 3.19	69.11 ± 3.20
12	41.48 ± 5.56	$\mathbf{75.04 \pm 4.08}$	31.86 ± 14.74	27.33 ± 5.93	47.30 ± 5.80	59.53 ± 5.65	63.67 ± 7.23	37.67 ± 8.26	62.81 ± 3.83
13	98.57 ± 1.43	$\mathbf{99.52 \pm 0.43}$	94.93 ± 3.40	75.85 ± 7.73	97.37 ± 1.00	92.10 ± 2.98	98.44 ± 0.90	95.28 ± 1.37	98.21 ± 0.72
14	55.70 ± 10.26	$\mathbf{67.91 \pm 3.48}$	4.63 ± 6.86	10.52 ± 3.63	62.02 ± 9.39	18.78 ± 8.03	24.93 ± 4.83	51.66 ± 7.10	40.04 ± 6.37
15	89.88 ± 2.82	82.76 ± 1.83	84.42 ± 9.66	81.62 ± 8.80	91.74 ± 2.90	91.97 ± 1.57	92.10 ± 1.48	89.08 ± 1.03	$\mathbf{92.57 \pm 1.53}$
16	65.20 ± 7.29	$\mathbf{88.91 \pm 2.60}$	26.15 ± 13.30	38.36 ± 6.14	74.56 ± 7.37	66.10 ± 3.13	68.05 ± 3.02	64.21 ± 2.94	65.80 ± 5.01
17	66.16 ± 8.98	$\mathbf{86.21 \pm 2.78}$	32.76 ± 9.76	37.90 ± 6.11	73.89 ± 10.03	68.06 ± 4.22	78.65 ± 6.74	79.20 ± 2.74	75.11 ± 3.64
18	84.35 ± 4.87	$\mathbf{96.88 \pm 1.00}$	88.17 ± 3.75	45.92 ± 15.18	92.46 ± 2.51	90.31 ± 2.23	89.58 ± 1.84	91.92 ± 3.77	90.37 ± 1.72
19	62.54 ± 7.00	63.64 ± 3.30	31.21 ± 19.94	59.60 ± 7.34	58.16 ± 7.01	$\mathbf{76.04 \pm 7.22}$	71.80 ± 5.90	71.21 ± 6.23	75.95 ± 4.02
20	61.92 ± 8.66	70.88 ± 4.03	47.27 ± 18.42	43.89 ± 6.39	69.70 ± 5.38	60.08 ± 6.26	66.92 ± 7.13	52.15 ± 6.07	$\mathbf{75.59 \pm 3.43}$
OA	47.94 ± 1.51	60.51 ± 1.55	31.87 ± 5.47	37.47 ± 2.69	54.82 ± 1.66	53.21 ± 3.27	58.24 ± 1.50	56.21 ± 1.28	$\mathbf{61.29 \pm 1.07}$
AA	60.23 ± 1.64	$\mathbf{74.26 \pm 0.86}$	39.26 ± 3.39	42.48 ± 3.51	64.79 ± 1.62	63.19 ± 2.36	67.83 ± 1.40	64.66 ± 0.71	70.45 ± 0.69
κ	43.62 ± 1.55	56.79 ± 1.60	27.59 ± 4.87	32.68 ± 2.75	50.60 ± 1.70	49.20 ± 3.37	54.37 ± 1.41	52.33 ± 1.22	$\mathbf{57.68 \pm 1.04}$

It is observed in Tables VI–IX that our model performs most stable with the lowest OA variance in the ten runs. This demonstrates that our model is less affected by the selection of training data, demonstrating the superior performance of our model when the training data is scarce. Although the SSFTT, GAHT, and Morphformer methods can capture the long-range spatial and spectral features, they still yield unsatisfactory performance. Mainly because these methods have difficulty in capturing local information without sufficient training data. On the contrary, our method shows a significant improvement over the other transformer-based methods by introducing discriminative spatial–spectral tokens and a spatial prior information-guided local transformer encoder to capture both local and global information.

2) Visual Evaluation: To visualize the classification performance of all the methods, we show the classification maps of different methods in Figs. 6–9. The results show that our method contains less salt and pepper noise and better aligns

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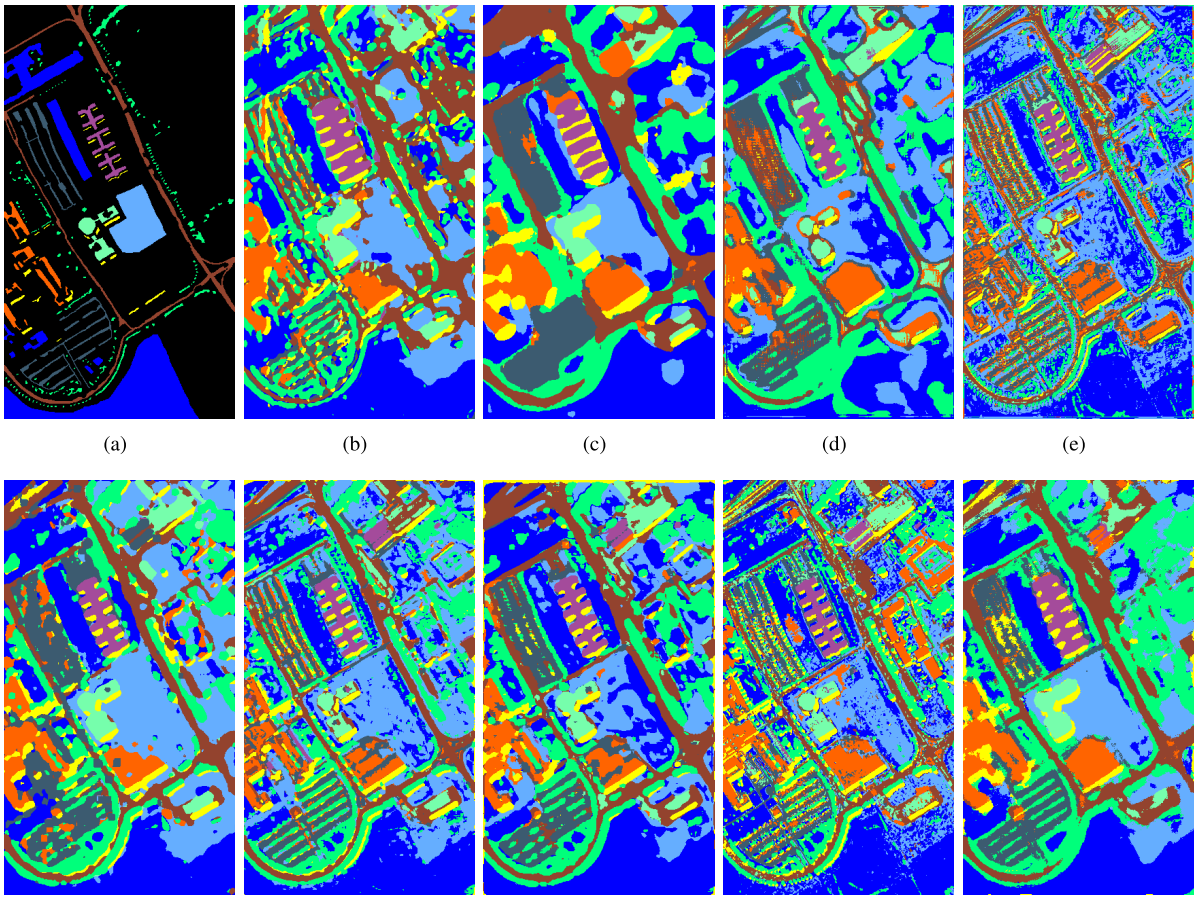


Fig. 6. *PaviaU* image. (a) Ground truth. (b) HybridSN (OA = 90.45%). (c) DFFN (OA = 93.67%). (d) HiT (OA = 71.14%). (e) Spectralformer (OA = 78.57%). (f) SSFTT (OA = 94.54%). (g) GAHT (OA = 92.91%). (h) Morphformer (OA = 92.94%). (i) GraphGST (OA = 93.04%). (j) Our proposed model (OA = 96.66%).

TABLE X

ABLATION ANALYSIS OF THE PROPOSED MODEL ON FOUR DATASETS

Schemes	Module			OA			
	MSST	DTE		Pavia	Houston	Longkou	SDFC
		Global branch	Local branch				
1	Linear	✓	✓	85.06	84.83	93.51	51.57
2	3D-CNN	✓	✓	91.57	84.22	96.00	52.48
3	✓	✓	×	92.12	88.03	97.39	57.22
4	✓	×	✓	90.00	89.99	96.86	55.48
5	✓	✓	✓	95.92	89.76	97.85	61.29

TABLE XI

INFLUENCE OF PATCH SIZE ON THE PERFORMANCE OF OUR MODEL ON FOUR DATASETS

Patch size	PaviaU	Houston2013	LongKou	SDFC
7×7	92.95	**89.76**	97.51	59.14
9×9	94.73	89.18	97.74	**61.29**
11×11	95.08	89.08	**97.85**	59.01
13×13	95.65	88.48	97.27	58.34
15×15	94.60	88.34	97.48	57.85
19×19	**95.92**	87.44	–	–
21×21	94.49	86.12	–	–
23×23	93.38	86.35	–	–

with the contours of land covers on all the datasets. For the PaviaU dataset, Fig. 6 shows that the result of our method is closer to the ground truth, especially in the "meadows" region at the bottom of the classification map. On the Houston2013 dataset, Fig. 7 shows that the classification map of our method has relatively fewer misclassification regions. On the LongKou dataset, Fig. 8 indicates that the classification map of our method closely matches the ground truth, especially on the boundaries of different objects. On the SDFC dataset, Fig. 9 shows that the classification map of our method has less salt and pepper noise, and is smoother in some large category areas (e.g., "Tree" and "Bare soil" areas on the left side of the classification map). Meanwhile, our method also shows clearer boundaries, such as the "road" region on the right side of the image with purple labeled pixels. The classification maps are consistent with the quantitative evaluations and further verify the superior performance of our method.

D. Ablation Study

MSST and the DTE are two essential components of the proposed model. To study the influence of MSST and DTE on the performance of our method, we replace MSST

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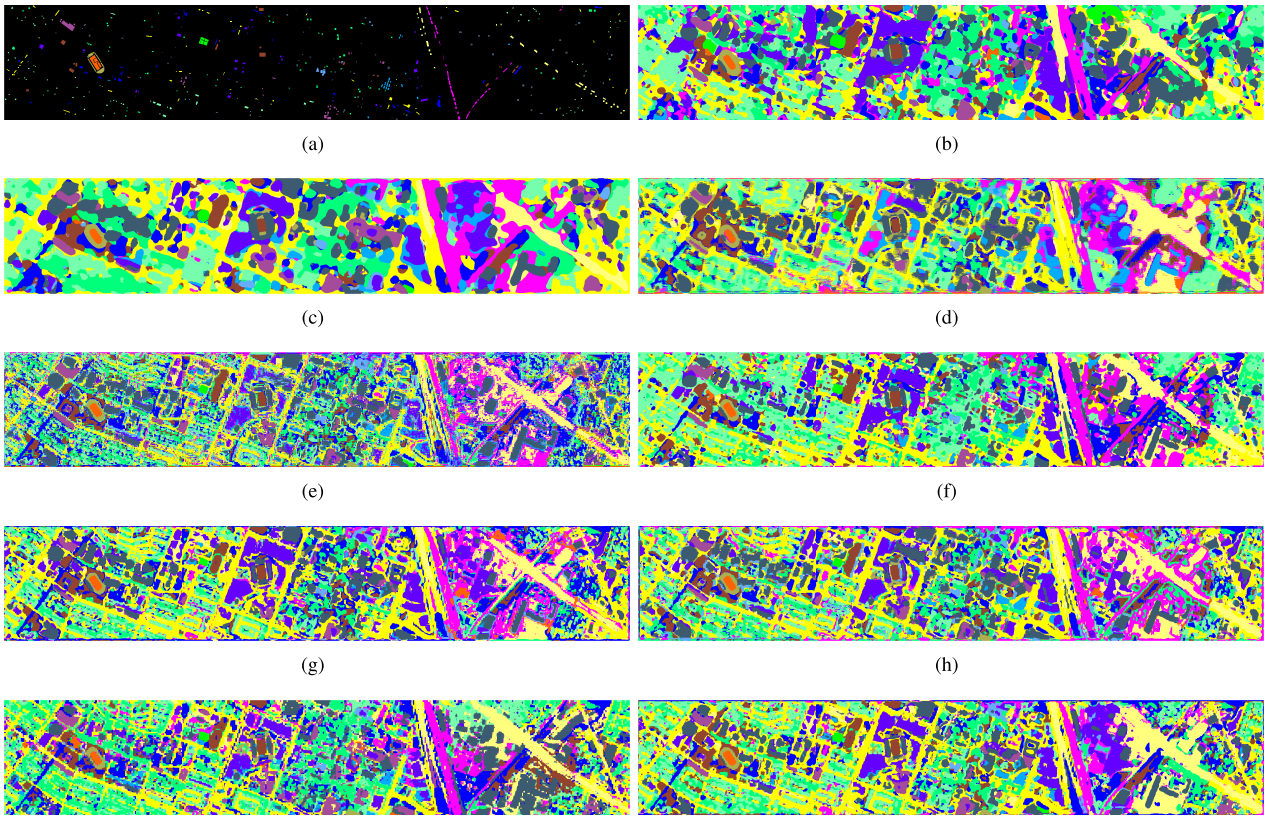


Fig. 7. *University of Houston* image. (a) Ground truth. (b) HybridSN (OA = 86.45%). (c) DFFN (OA = 86.20%). (d) HiT (OA = 68.78%). (e) Spectralformer (OA = 82.40%). (f) SSFTT (OA = 89.30%). (g) GAHT (OA = 86.37%). (h) Morphformer (OA = 89.16%). (i) GraphGST (OA = 90.50%). (j) Our proposed model (OA = 90.12%).

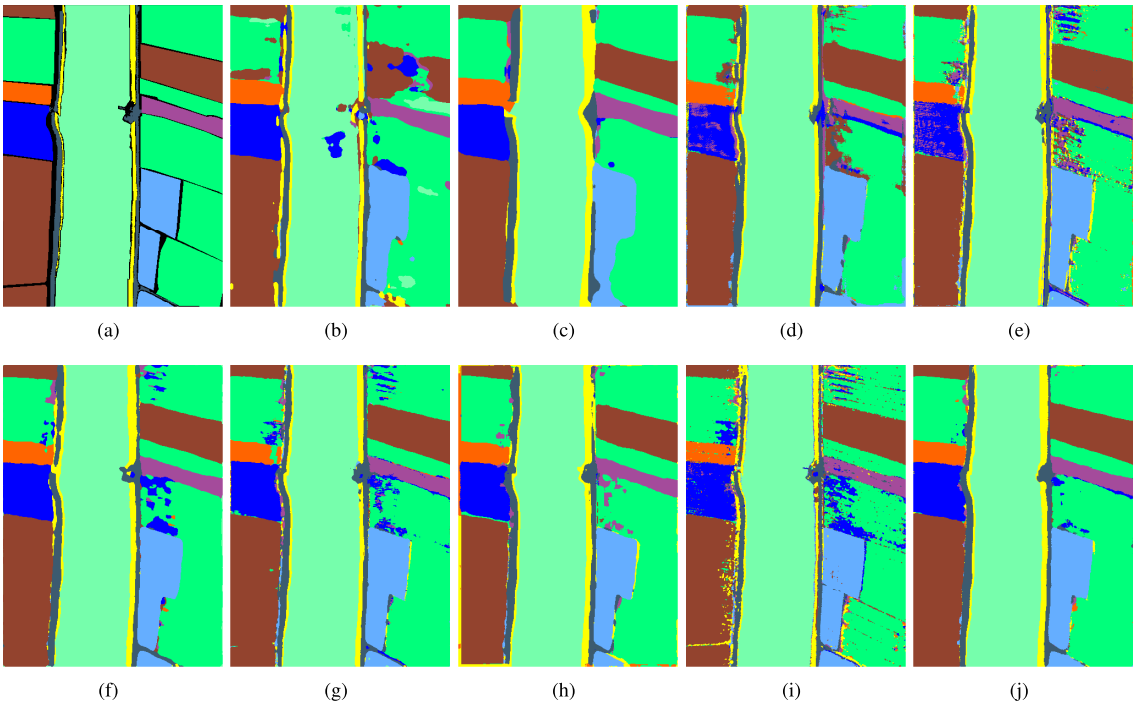


Fig. 8. *WHU-Hi-LongKou* image. (a) Ground truth. (b) HybridSN (OA = 91.38%). (c) DFFN (OA = 95.77%). (d) HiT (OA = 92.45%). (e) Spectralformer (OA = 92.53%). (f) SSFTT (OA = 96.10%). (g) GAHT (OA = 96.69%). (h) Morphformer (OA = 95.74%). (i) GraphGST (OA = 95.71%). (j) Our proposed model (OA = 98.24%).

with a regular linear projection layer and a 3D-CNN block, respectively, to verify the effectiveness of the MSST module. The 3D-CNN block consists of a 3-D convolutional layer, a Relu activation function, and a batch normalization layer. The results of the ablation experiments are shown in Table X. It is observed that scheme 5, which employs both the MSST

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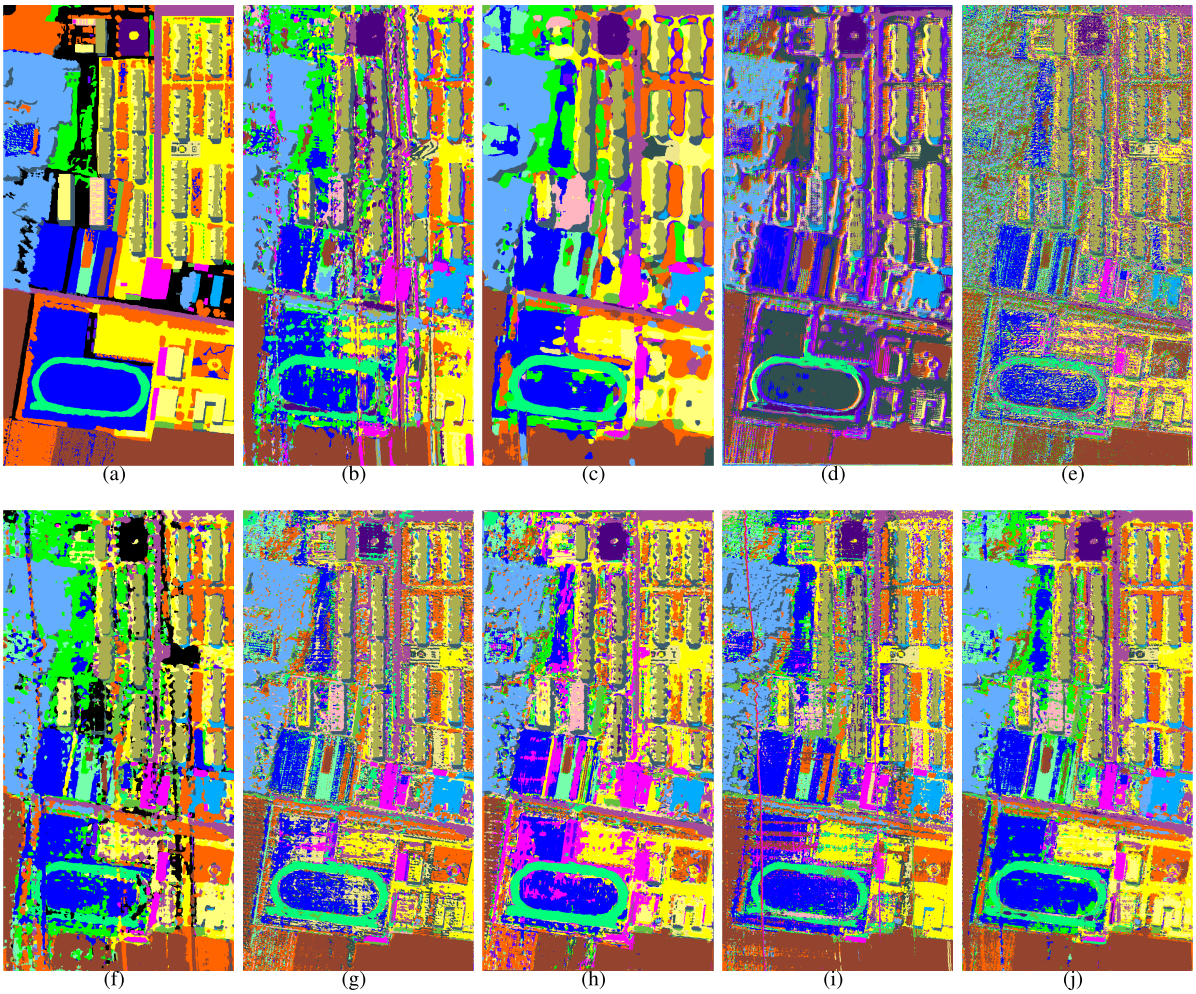


Fig. 9. *SDFC* image. (a) Ground truth. (b) HybridSN (OA = 47.92%). (c) DFFN (OA = 60.53%). (d) HiT (OA = 32.95%). (e) Spectralformer (OA = 37.88%). (f) SSFTT (OA = 54.70%). (g) GAHT (OA = 53.26%). (h) Morphformer (OA = 58.44%). (i) GraphGST (OA = 56.32%). (j) Our proposed model (OA = 61.29%).

TABLE XII
FLOPS AND PARAMETERS OF DIFFERENT MODELS ON THE FOUR DATASETS

Datasets	Metrics	HybridSN	DFFN	HiT	Spectralformer	SSFTT	GAHT	Morphformer	GraphGST	Proposed
PaviaU	Flops (M)	7272.35	11317.30	74091.92	1766.41	2052.36	31035.55	4118.40	**22.18**	3869.61
	Params (K)	2369.66	491.69	48218.22	231.02	153.22	927.11	226.86	408.58	**75.35**
Houston	Flops (M)	390.09	3798.96	17184.38	2358.55	240.43	5530.44	1156.01	**139.96**	474.03
	Params (K)	403.58	492.80	51175.17	241.65	153.22	973.58	232.69	4023.23	**70.79**
LongKou	Flops (M)	1024.87	3798.93	60172.16	6779.51	459.19	9742.36	3115.51	**133.49**	1285.20
	Params (K)	534.66	492.41	66636.11	571.88	153.22	1118.35	333.10	3040.02	**135.47**
SDFC	Flops (M)	390.09	2543.14	11042.98	700.84	240.43	5052.70	623.39	**78.28**	354.77
	Params (K)	403.58	493.12	46172.26	134.97	153.22	881.75	184.05	1759.20	**47.63**

module and DTE, yields the best performance. Compared with Scheme 1 and Scheme 2, Scheme 5 obtains a significant OA improvement in all four datasets, which indicates the importance of our tokenization module.

We also investigate the influence of each branch of DTE on the performance of our model. Different schemes are shown in Table X. Compared with Scheme 3, Scheme 5 obtains better performance on all four data sets in terms of OA. This verifies the benefit of our local branch. Compared with Scheme 4, Scheme 5 obtains an improved OA on PaviaU, Longkou, and SDFC, which demonstrates the effectiveness of the global branch.

E. Parameter Analysis

1) Influence of the Patch Size: We investigate the influence of the input patch size on the performance of our model

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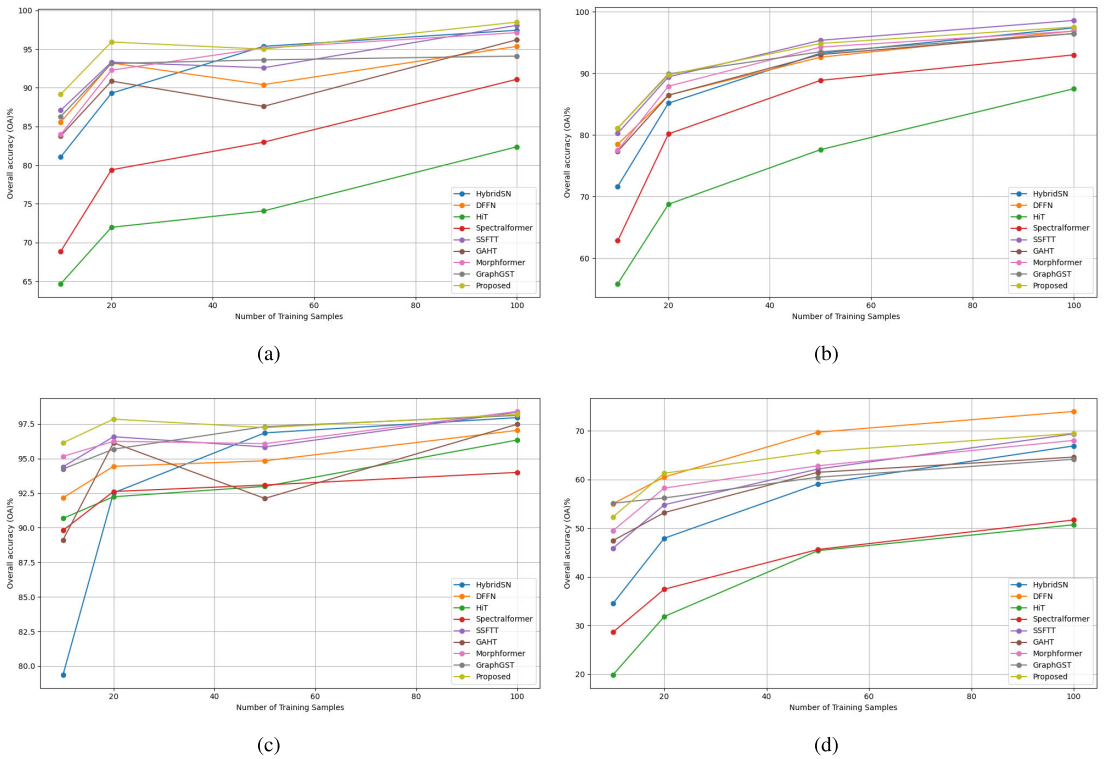


Fig. 10. Influence of the number of training samples on the four datasets. (a) PaviaU. (b) Houston2013. (c) LongKou. (d) SDFC.

on four datasets, including PaviaU, Houston2013, LongKou, and SDFC. We set the patch size to 7×7, 9×9, 11×11, 13×13, 15×15, 19×19, 21×21, and 23×23. The results are reported in Table XI. Note that due to the out-of-memory problem on LongKou and SDFC, the results with patch sizes of 19×19, 21×21 and 23×23 are not reported. The results show that the optimal patch size is 19×19, 7×7, 11×11, and 9×9 for PaviaU, Houston2013, LongKou, and SDFC, respectively, which depends on the type of landcovers. It should be noted that with different patch sizes our model performs better than the state-of-the-art in most cases, which is evidenced by the results in Tables VI–IX and XI.

2) Influence of the Number of Training Samples: In order to evaluate the adaptability of the models to limited training data, we randomly select 10, 20, 50, and 100 samples per class from the four datasets as training sets. The results are shown in Fig. 10. When the number of training samples is small (ten samples), there is a large accuracy gap between different methods. However, the gap gets smaller with more training data. In general, more training data often leads to improved performance. Our method performs the best in most cases, especially when using a few training data, i.e., 10 and 20 samples per class. We can also observe that our method performs well in all training sample size conditions, especially when the number of training samples per class is small, e.g., 10 and 20, showing superior performance on the four datasets.

F. Efficiency Analysis

We evaluate the floating point operations per second (Flops) and the number of parameters of different methods on four datasets, and the results are shown in Table XII. Due to the utilization of a large number of convolutional layers, CNN-based methods (HybridSN and DFFN) exhibit a relatively large number of parameters. About 2370 and 492 K learnable parameters in HybridSN and DFFN on the PaviaU dataset indicate that CNN architectures usually have more parameters of convolutional kernels to learn features of HSIs. For transformer-based architectures, HiT exhibits the largest number of parameters on all four datasets, since transformer-based methods usually require more parameters to handle complex feature relationships. GraphGST has the lowest Flops among all methods; however, it contains more parameters to be learned compared with our method. In general, our method achieves a good balance between computational efficiency and model complexity. This highlights the strong, lightweight nature of our model while ensuring competitive performance in HSI classification.

V. Conclusion

In this article, we propose a novel HSI classification method based on the locally enhanced transformer network. Our method consists of the MSST module and the DTE module. The former is developed to extract discriminative spectral-spatial information of HSI for effective tokenization, which alleviates the problem of information loss during tokenization. The dual-branch transformer module combines a global ViT encoder and a locally enhanced ViT encoder guided by super-pixel segmentation. This integration enhances both global and local feature extraction capabilities. Due to the employed single layer of CNN and transformer encoder in our network, our model is lightweight and contains fewer

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parameters to be optimized, which favors applications when only a few training data are available. Experimental results on benchmark datasets demonstrate the superior performance of our method compared to the state-of-the-art.

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REFERENCES

[1] S. Huang, H. Zhang, and A. Pižurica, "Subspace clustering for hyperspectral images via dictionary learning with adaptive regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524017.

[2] L. Wang, B. Zuo, Y. Le, Y. Chen, and J. Li, "Penetrating remote sensing: Next-generation remote sensing for transparent Earth," *Innov.*, vol. 4, no. 6, Nov. 2023, Art. no. 100519.

[3] S. Peyghambari and Y. Zhang, "Hyperspectral remote sensing in lithological mapping, mineral exploration, and environmental geology: An updated review," *J. Appl. Remote Sens.*, vol. 15, no. 3, Jul. 2021, Art. no. 031501.

[4] M. B. Stuart, A. J. S. McGonigle, and J. R. Willmott, "Hyperspectral imaging in environmental monitoring: A review of recent developments and technological advances in compact field deployable systems," *Sensors*, vol. 19, no. 14, p. 3071, Jul. 2019.

[5] M. Lv et al., "Spatial–spectral density peaks-based discriminant analysis for membranous nephropathy classification using microscopic hyperspectral images," *IEEE J. Biomed. Health Informat.*, vol. 25, no. 8, pp. 3041–3051, Aug. 2021.

[6] E. Blanzieri and F. Melgani, "Nearest neighbor classification of remote sensing images with the maximal margin principle," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 6, pp. 1804–1811, Jun. 2008.

[7] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[8] Y. Zhang, G. Cao, X. Li, and B. Wang, "Cascaded random forest for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1082–1094, Apr. 2018.

[9] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Classification for hyperspectral imagery based on sparse representation," in *Proc. 2nd Workshop Hyperspectral Image Signal Process., Evol. Remote Sens.*, Reykjavik, Iceland: IEEE, Jun. 2010, pp. 1–4.

[10] J. Peng, L. Li, and Y. Y. Tang, "Maximum likelihood estimation-based joint sparse representation for the classification of hyperspectral remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 6, pp. 1790–1802, Jun. 2019.

[11] J. Xia, N. Falco, J. A. Benediktsson, P. Du, and J. Chanussot, "Hyperspectral image classification with rotation random forest via KPCA," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 4, pp. 1601–1609, Apr. 2017.

[12] W. Li, S. Prasad, J. E. Fowler, and L. M. Bruce, "Locality-preserving discriminant analysis in kernel-induced feature spaces for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 5, pp. 894–898, Sep. 2011.

[13] X. Zhang, Z. Gao, L. Jiao, and H. Zhou, "Multifeature hyperspectral image classification with local and nonlocal spatial information via Markov random field in semantic space," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1409–1424, Mar. 2018.

[14] Y. Ren, L. Liao, S. J. Maybank, Y. Zhang, and X. Liu, "Hyperspectral image spectral–spatial feature extraction via tensor principal component analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 9, pp. 1431–1435, Sep. 2017.

[15] J. Peng, W. Sun, and Q. Du, "Self-paced joint sparse representation for the classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1183–1194, Feb. 2019.

[16] S. Huang, H. Zhang, J. Xue, and A. Pižurica, "Heterogeneous regularization-based tensor subspace clustering for hyperspectral band selection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 9259–9273, Nov. 2023.

[17] S. Huang, H. Zhang, and A. Pižurica, "A structural subspace clustering approach for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5509515.

[18] G. Camps-Valls, L. Gomez-Chova, J. Munoz-Mari, J. Vila-Frances, and J. Calpe-Maravilla, "Composite kernels for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 1, pp. 93–97, Jan. 2006.

[19] P. Sadeghi-Tehran, N. Virlet, and M. J. Hawkesford, "A neural network method for classification of sunlit and shaded components of wheat canopies in the field using high-resolution hyperspectral imagery," *Remote Sens.*, vol. 13, no. 5, p. 898, Feb. 2021.

[20] S. Jia, B. Deng, J. Zhu, X. Jia, and Q. Li, "Local binary pattern-based hyperspectral image classification with superpixel guidance," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 749–759, Feb. 2018.

[21] J. Liu, Z. Wu, J. Li, A. Plaza, and Y. Yuan, "Probabilistic-kernel collaborative representation for spatial–spectral hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 2371–2384, Apr. 2016.

[22] M. Fauvel, J. Chanussot, and J. A. Benediktsson, "Kernel principal component analysis for the classification of hyperspectral remote sensing data over urban areas," *EURASIP J. Adv. Signal Process.*, vol. 2009, no. 1, pp. 1–14, Dec. 2009.

[23] M. Fauvel, J. A. Benediktsson, J. Chanussot, and J. R. Sveinsson, "Spectral and spatial classification of hyperspectral data using SVMs and morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 11, pp. 3804–3814, Nov. 2008.

[24] C. Zhao, W. Zhu, and S. Feng, "Superpixel guided deformable convolution network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3838–3851, 2022.

[25] R. Liu, T. Luo, S. Huang, Y. Wu, Z. Jiang, and H. Zhang, "CrossMatch: Cross-view matching for semi-supervised remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5650515.

[26] L. Liang, S. Zhang, and J. Li, "Multiscale DenseNet meets with bi-RNN for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 5401–5415, 2022.

[27] F. Zhang, J. Bai, J. Zhang, Z. Xiao, and C. Pei, "An optimized training method for GAN-based hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 10, pp. 1791–1795, Oct. 2021.

[28] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel-and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2020.

[29] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jan. 2015.

[30] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[31] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[32] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[33] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[34] C. Zhang, G. Li, and S. Du, "Multi-scale dense networks for hyperspectral remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9201–9222, Nov. 2019.

[35] S. Dong, W. Feng, Y. Quan, G. Dauphin, L. Gao, and M. Xing, "Deep ensemble CNN method based on sample expansion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531815.

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[36] H. Sun, X. Zheng, X. Lu, and S. Wu, "Spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3232–3245, May 2020.

[37] Z. Lu, B. Xu, L. Sun, T. Zhan, and S. Tang, "3-D channel and spatial attention based multiscale spatial–spectral residual network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4311–4324, 2020.

[38] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.

[39] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, Jan. 2021.

[40] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2020, pp. 1–21.

[41] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.

[42] E. Xie, N. Chen, J. Peng, W. Sun, Q. Du, and X. You, "Semantic and spatial–spectral feature fusion transformer network for the classification of hyperspectral image," *CAAI Trans. Intell. Technol.*, vol. 8, no. 4, pp. 1308–1322, Feb. 2023.

[43] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[44] F. Zhao, S. Li, J. Zhang, and H. Liu, "Convolution transformer fusion splicing network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[45] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.

[46] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.

[47] Y. Peng, Y. Zhang, B. Tu, Q. Li, and W. Li, "Spatial–spectral transformer with cross-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537415.

[48] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[49] W. Zeng et al., "Microscopic hyperspectral image classification based on fusion transformer with parallel CNN," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 6, pp. 2910–2921, Jun. 2023.

[50] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jan. 2022, Art. no. 5522214.

[51] H. Yu, Z. Xu, K. Zheng, D. Hong, H. Yang, and M. Song, "MSTNet: A multilevel spectral–spatial transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532513.

[52] P. Ghosh, S. K. Roy, B. Koirala, B. Rasti, and P. Scheunders, "Hyperspectral unmixing using transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535116.

[53] S. Jia, Z. Min, and X. Fu, "Multiscale spatial–spectral transformer network for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 96, pp. 117–129, Aug. 2023.

[54] Z. Peng et al., "Conformer: Local features coupling global representations for visual recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 367–376.

[55] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Hyperspectral image transformer classification networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528715.

[56] J. Zhang, Z. Meng, F. Zhao, H. Liu, and Z. Chang, "Convolution transformer mixer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[57] F. Xu, G. Zhang, C. Song, H. Wang, and S. Mei, "Multiscale and cross-level attention learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501615.

[58] A. Yang, M. Li, Y. Ding, D. Hong, Y. Lv, and Y. He, "GTFN: GCN and transformer fusion network with spatial–spectral features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 6600115.

[59] M. Jiang et al., "GraphGST: Graph generative structure-aware transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504016.

[60] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral image classification using groupwise separable convolutional vision transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511817.

[61] Y. Cai, Z. Zhang, Z. Cai, X. Liu, X. Jiang, and Q. Yan, "Graph convolutional subspace clustering: A robust subspace clustering framework for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4191–4202, May 2021.

[62] T. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. 5th Int. Conf. Learn. Represent. (ICLR)*, Toulon, France, Jan. 2016, pp. 1–14.

[63] R. Fan, J. Li, W. Song, W. Han, J. Yan, and L. Wang, "Urban informal settlements classification via a transformer-based spatial–temporal fusion network using multimodal remote sensing and time-series human activity data," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 111, Jul. 2022, Art. no. 102831.

[64] Y. Xu, Y. Xu, H. Jiao, Z. Gao, and L. Zhang, "S^3ANet: Spatial–spectral self-attention learning network for defending against adversarial attacks in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512913.

[65] T. Gao, C.-Z. Xu, L. Zhang, and H. Kong, "GSB: Group superposition binarization for vision transformer with limited training samples," *Neural Netw.*, vol. 172, Apr. 2024, Art. no. 106133.

[66] M.-Y. Liu, O. Tuzel, S. Ramalingam, and R. Chellappa, "Entropy rate superpixel segmentation," in *Proc. CVPR*, Jun. 2011, pp. 2097–2104.

[67] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-Hi: UAV-borne hyperspectral with high spatial resolution (H2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.

[68] L. Sun, J. Zhang, J. Li, Y. Wang, and D. Zeng, "SDFC dataset: A large-scale benchmark dataset for hyperspectral image classification," *Opt. Quantum Electron.*, vol. 55, no. 2, p. 173, Feb. 2023.

[69] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.



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